PROJECT DOSSIER: BHARAT HERITAGE PORTAL

An Indigenous Digital Infrastructure for Indian Culture & History

Prepared By: Technical Team

Version: 2.0.0 (Strict Compliance Build)

Date: February 16, 2026

TABLE OF CONTENTS

1. **Executive Summary**
2. **Product Vision & Strategic Alignment**
3. **Current Progress Status (Phase 1)**
4. **Requirements from Stakeholders (NCERT / Ministry)**
5. **Technical Architecture (The "Sovereign Stack")**
6. **Detailed Feature Specifications**
7. **Content & Asset Standards**
8. **Future Roadmap**

1. EXECUTIVE SUMMARY

The **Bharat Heritage Portal** is a cutting-edge, AI-driven educational platform designed to showcase India's 5,000+ year history, geography, and art forms. Unlike existing platforms that rely on foreign mapping APIs (Google Maps/OpenStreetMap), this initiative is **100% indigenous**, utilizing official **Survey of India (SOI)** vector data to ensure territorial integrity.

The platform is engineered for **Zero-Latency Performance**, enabling students in low-bandwidth rural areas (e.g., Kalahandi) to experience high-fidelity cultural content without buffering. It is a direct technical response to the Prime Minister's vision of **"Digital Sovereignty"** and the **NEP 2020** mandate for experiential learning.

2. PRODUCT VISION & STRATEGIC ALIGNMENT

2.1. The Core Problem

Currently, Indian students learn geography and history through static textbooks or foreign platforms (Google Arts & Culture) that:

1. Do not always respect India's official territorial boundaries.
2. Host data on foreign servers, creating data sovereignty risks.
3. Lack a unified "Bharatiya" narrative, often presenting history in a fragmented manner.

2.2. The Solution: "Digital Sovereignty"

- **Indigenous Tech Stack:** Built entirely on open-source technologies by Indian technocrats.
- **Compliance First:** Hard-coded adherence to Survey of India map boundaries.
- **Indian Knowledge Systems (IKS):** Prioritizes Indian historical eras (e.g., Mauryan, Gupta, Maratha) over colonial categorizations.

3. CURRENT PROGRESS STATUS (PHASE 1)

Reporting Period: Feb 11, 2026 – Feb 14, 2026

The development team has successfully established the architectural foundation and core engines.

- **Architecture:** Finalized "Local-First" JSON architecture (No external API calls).
- **Map Engine:** Functional SVG Map with smooth "Click-to-Zoom" transitions (D3.js).
- **Timeline Engine:** Functional Horizontal Scroll history viewer.
- **Assets:** Asset pipelines for Audio (MP3) and Images (WebP) are active.
- **Compliance:** Map border verification protocol is implemented.

4. REQUIREMENTS FROM STAKEHOLDERS (NCERT / MINISTRY)

To move from **Prototype** to **National Pilot**, we require the following specific support from the Ministry of Culture or NCERT (CIET):

4.1. Content Vetting & Validation (Crucial)

We request the appointment of a **Nodal Officer** or a **Subject Matter Expert (SME)** Committee to:

- **Validate Map Boundaries:** Officially certify that our vector maps align with the Survey of India's latest mandates to prevent legal liabilities.
- **Vet Historical Text:** Review the 150+ timeline events and state cultural descriptions for factual accuracy and cultural sensitivity.
- **Approve "Indian Knowledge System" Categorization:** Ensure our era groupings (e.g., "Golden Age," "Colonial Resistance") align with the latest NCERT curriculum framework.

4.2. Infrastructure Support

- **Hosting:** We request NIC cloud credits or a sub-domain allocation (e.g., heritage.education.gov.in) to host the platform on government servers for credibility.
- **DIKSHA Integration:** Access to the **DIKSHA API** sandbox to test embedding our tool within the national education portal.

4.3. Pilot Authorization

- **School Access:** An official authorization letter to conduct a **Pilot Deployment** in 10-50 government schools (specifically in aspirational districts like Kalahandi, Odisha).

5. TECHNICAL ARCHITECTURE (THE "SOVEREIGN STACK")

5.1. Technology Stack

- **Frontend:** React 18, TypeScript, Vite (for performance).
- **Styling:** Tailwind CSS (for "Dark Heritage" design system).
- **Animation:** Framer Motion (for cinematic UI transitions).
- **Visualization:** D3.js (for Vector Map projection).
- **Data Layer:** Local JSON (No Database/Backend required for Phase 1).

5.2. Key Constraints

- **No Foreign APIs:** Strictly forbids calls to Google Maps, Mapbox, or OpenStreetMap.
- **Offline Capable:** Designed to be easily packaged as a Progressive Web App (PWA) for offline tablets.
- **Low Bandwidth:** All assets are compressed (WebP/MP3) to ensure load times < 1.5s on 4G networks.

6. DETAILED FEATURE SPECIFICATIONS

6.1. Home Page (The Interactive Atlas)

- **Visual:** Full-screen "Dark Mode" India Map.
- **Interaction:**
  - **Hover:** Regions glow with Saffron accent color.
  - **Click:** Camera smooth-zooms into the state; "Swoosh" sound plays.
- **Dashboard Overlay:** A side panel slides in displaying:
  - **Facts:** Capital, Language, Area.
  - **Culture:** Brief on Dance, Music, and Festivals.
  - **Monuments:** Scrollable list of key heritage sites.

6.2. Timeline Page (The Chronicle)

- **Visual:** A continuous horizontal "River of Time."
- **Interaction:**
  - **Scroll:** Vertical mouse wheel translates to horizontal history scrolling.
  - **Event Nodes:** Glowing dots representing key events (e.g., 1947 Independence).
  - **Detail Cards:** Clicking a node expands a glass-card with historical context.
- **Structure:** Events are grouped by Eras (Ancient, Medieval, Modern) with visual background shifts.

6.3. Global Components

- **Navigation Dock:** A floating glass capsule at the bottom center for switching views.
- **Audio Controller:** A global mute toggle for sound effects and ambient music.
- **Companion Widget:** A static avatar providing context-aware hints (e.g., "Click on Rajasthan to see Forts").

7. CONTENT & ASSET STANDARDS

7.1. File Structure Strategy

```
Plaintext
src/
 ├── data/
 │   ├── geo-india-official.json   # OFFICIAL TOPOJSON (SOI Compliant)
 │   ├── states-content.json       # Cultural Data Database
 │   └── timeline-events.json      # Historical Data Database
 ├── assets/
 │   ├── audio/                # .mp3 (128kbps)
 │   ├── covers/               # .webp (State Cover Images)
 │   └── monuments/            # .webp (Monument Images)
```

7.2. Data Standards

- **Images:** WebP Format Only. Max width 800px.
- **Audio:** MP3 Format Only. 128kbps bitrate.
- **Maps:** TopoJSON format validated against official GOI maps.

8. FUTURE ROADMAP (PHASE 2 & 3)

- **Phase 2 (Q3 2026):**
  - **Gamification:** "Heritage Quiz" unlocked after exploring 5 states.
  - **Language Support:** Localization into Hindi, Odia, and Tamil.
  - **Virtual Tours:** 360-degree image viewer for select monuments (e.g., Konark Sun Temple).
- **Phase 3 (2027):**
  - **AR Integration:** Augmented Reality viewer for classroom tablets.
  - **Teacher Dashboard:** Analytics for teachers to track student exploration.

[End of Document]

Report Prepared By: Development Team

Sign-off: Project Lead



Abhishek Singh, project head